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Non-toxic protease having improved productivity

Abstract

The present invention relates to a mutated non-toxic protease in which the amino acid cysteine (Cys) at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid other than cysteine. According to the present invention, it is possible to recover a refolded non-toxic protease from an insoluble fraction from which the non-toxic protease was almost impossible to recover in the prior art, and thus it is possible to produce the non-toxic protease with significantly improved productivity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a United States national phase under 35 USC § 371 of International Patent Application No. PCT/KR2020/005623 filed Apr. 28, 2020, which in turn claims priority under 35 USC § 119 of Korean Patent Application No. 10-2019-0069259 filed Jun. 12, 2019. The disclosures of all such applications are hereby incorporated herein by reference in their respective entireties, for all purposes.
REFERENCE TO SEQUENCE LISTING SUBMITTED VIA EFS-WEB
(2) This application includes an electronically submitted sequence listing in .txt format. The .txt file contains a sequence listing entitled “607_SeqListing_ST25.txt” created on Dec. 7, 2021 and is 96,478 bytes in size. The sequence listing contained in this .txt file is part of the specification and is hereby incorporated by reference herein in its entirety.
TECHNICAL FIELD
(3) The present invention relates to a non-toxic protease with improved productivity, and more particularly, to a non-toxic protease which is produced with improved productivity as a result of inhibiting irreversible protein aggregation and improving refolding efficiency through introduction of a point mutation.
BACKGROUND ART
(4) Some of the best known examples of substances that act on nerve cells by incapacitating the secretory function of cellular physiological substances in target cells include clostridial neurotoxins (e.g., botulinum neurotoxins commercially available under trade names such as Dysport™, Neurobloc™, and Botox™). The non-toxic protease domains constituting these substances are zinc-endopeptidases which are a well-known group of proteases that act on target cells by incapacitating the secretory function of cellular physiological substances. Interestingly, the non-toxic protease domains alone do not kill the target cells on which they act.
(5) The non-toxic protease domains act by proteolytically cleaving SNARE proteins (e.g., intracellular transport proteins known as SNAP-25, VAMP, or syntaxin) (see Gerald K (2002) “Cell and Molecular Biology” (4th edition) John Wiley & Sons, Inc.). The acronym “SNARE” derives

from Soluble NSF Attachment Receptor, where NSF means N-ethylmaleimide—Sensitive Factor. SNARE proteins are integral to intracellular vesicle formation, and thus to secretion of molecules through vesicle transport from a cell. Accordingly, once delivered to a desired target cell, the non-toxic protease is capable of inhibiting cellular secretion from the target cell.

(6) Clostridial neurotoxins represent a major group of non-toxic toxin molecules and comprise two polypeptide chains joined together by a disulfide bond. The two chains are termed the heavy chain (H-chain) having a molecular weight of about 100 kDa and the light chain (L-chain) having a molecular weight of about 50 kDa. It is the L-chain, which possesses a protease function and exhibits a high substrate specificity for vesicle and/or plasma membrane associated (SNARE) proteins involved in the exocytic process (e.g., synaptobrevin, syntaxin or SNAP-25). These substrates are important components of the neurosecretory machinery (Korean Patent Application Publication No. 10-2014-0036239).

(7) Non-toxic proteases can temporarily inhibit muscle contraction in vivo, and thus they are used as bio-drugs for the treatment of some Dystonia or muscle contraction disorders. That is, non-toxic proteases inhibit muscle contraction by temporarily inhibiting the release of acetylcholine. Recently, non-toxic proteases have been successfully used as injectable biological drugs in the treatment of various muscle contraction disorders, such as blepharospasm, migraine headache, reduction of facial wrinkles, etc. (Mauskop A. The use of Botulinum Toxin in the treatment of headaches. Curr Pain Headache Rep 2002; 6:320-3; Mauskop A. Botulinum toxin in headache treatment: The end of the road? Cephalalgia 2007; 27:468; Verheyden J, Blitzer A. Other noncosmetic uses of Botox. Dis Mon 2002; 48:357-66; Dhaked R K, Singh M K, Singh P, Gupta P. Botulinum toxin: Bioweapon & magic drug. Indian J Med Res 2010; 132:489-503; Turton K, Chaddock J A, Acharya K R. Botulinum and tetanus neurotoxins: structure, function and therapeutic utility. Trends Biochem Sci 2002; 27:552-8; Masuyer G, Chaddock J A, Foster K A, Acharya K R. Engineered botulinum neurotoxins as new therapeutics. Annu Rev Pharmacol Toxicol 2014; 54:27-51).

(8) Non-toxic proteases can be used for various purposes as described above and have high commercial potential, and thus attempts have been made to mass-produce non-toxic proteases using recombinant microorganisms. However, a significant portion of a recombinant non-toxic protease expressed in a recombinant microorganism transformed with a recombinant expression vector tends to aggregate in the form of inclusion bodies in an insoluble fraction depending on the composition and structure of the protein. In this case, the protein is dissolved as much as possible in a solution of urea, guanidium chloride, a surfactant or a reducing agent, and then purified through centrifugation, dialysis, ion exchange, hydrophobic interaction, reverse phase and gel filtration HPLC, various chromatography columns packed with an appropriate type of resin, etc. The solution used in this complex purification process changes the structure of the protein, resulting in loss of the activity of the non-toxic protease, and even when a complex additional process of refolding the purified protein to restore the protease activity is introduced, a problem arises in that a problem that most of the inclusion bodies are not recovered due to low restoration efficiency (Saffarian, BIOENGINEERED 2016, VOL. 7, NO. 6, 478-483). This refolding process is generally performed by diluting or dialyzing the protein in a buffer solution free of a reducing/denaturing agent depending on the size of the protein to induce disulfide bonds between cysteine amino acids inside the protein through oxidation, and then separating soluble and insoluble fractions by filtration or centrifugation, thereby recovering only a protein whose unique three-dimensional structure has been restored. Even when this complicated process is performed, the recovery rate of the refolded non-toxic protease from the insoluble fraction is quite low (Cabrita, 2004, Biotechnology Annual Review, 10:31-50; Clark, Current Opinion in Biotechnology 1998, 9:157-163; Yamaguchi, Biomolecules 2014, 4, 235-251).

(9) Accordingly, the present inventors have made extensive efforts to effectively recover a refolded non-toxic protease even from insoluble fractions in which most of non-toxic proteases are present

but which are almost impossible to recover due to their irreversible aggregation into inclusion bodies, in a process of producing a non-toxic protease using a recombinant microorganism. As a result, the present inventors have found that, when a point mutation is induced in some amino acids of a non-toxic protease, it is possible to recover a non-toxic protease that reversibly aggregates into inclusion bodies, is effectively refolded even in insoluble fractions, and has activity, and the production of the non-toxic protease may be maximized, thereby completing the present invention.

SUMMARY OF THE INVENTION

(10) An object of the present invention is to provide a mutated non-toxic protease which is produced in improved yield using a recombinant microorganism, and a method for producing the same.

(11) To achieve the above object, the present invention provides a mutated non-toxic protease in which the amino acid cysteine at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid other than cysteine.

(12) The present invention also provides a fusion non-toxic proteinase in which the mutated non-toxic protease is fused to any one or more peptides selected from the group consisting of: i) a cell penetrating peptide; ii) a belt domain fragment peptide; and iii) a cell targeting peptide.

(13) The present invention also provides a fusion non-toxic proteinase in which the mutated non-toxic protease is fused with any one or more peptides selected from the group consisting of: i) a yeast secretion signal peptide; ii) a cell penetrating peptide; and ii) a cell targeting peptide.

(14) The present invention also provides a mutant gene encoding the mutated non-toxic protease.

(15) The present invention also provides a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of i) a nucleic acid encoding a cell penetrating peptide; ii) a nucleic acid encoding a belt domain fragment peptide; and iii) a nucleic acid encoding a cell targeting peptide.

(16) The present invention also provides a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of i) a nucleic acid encoding a yeast secretion signal peptide; ii) a nucleic acid encoding a cell penetrating peptide; and iii) a nucleic acid encoding a cell targeting peptide.

(17) The present invention also provides a recombinant vector into which the mutant gene has been introduced.

(18) The present invention also provides a recombinant microorganism into which the recombinant vector has been introduced.

(19) The present invention also provides a method for producing a mutated non-toxic protease or a fusion non-toxic protease, the method comprising steps of: (a) producing the mutated non-toxic protease or the fusion non-toxic protease by culturing the recombinant microorganism; (b) disrupting the recombinant microorganism; and (c) purifying the mutated non-toxic protease or the fusion non-toxic protease from the disrupted recombinant microorganism.

(20) The present invention also provides a pharmaceutical composition for treating Dystonia containing the mutated or fusion non-toxic protease as an active ingredient.

(21) The present invention also provides a hyaluronic acid microneedle patch containing the mutated or fusion non-toxic protease.

(22) The present invention also provides a method for alleviating or treating symptoms of Dystonia, the method comprising a step of administering the mutated or fusion non-toxic protease to a subject in need of alleviation or treatment of symptoms of Dystonia.

(23) The present invention also provides the use of the mutated or fusion non-toxic protease for the manufacture of a medicament for use in the alleviation or treatment of symptoms of Dystonia.

(24) The present invention also provides the above mutated or fusion non-toxic protease for use in a method for alleviating or treating symptoms of Dystonia.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** schematically shows the structural difference between a mutated non-toxic protease according to the present invention and a wild-type non-toxic protease.

(2) FIG. **2** schematically shows the principle that irreversible insoluble inclusion bodies are formed as a result of expression of the wild-type non-toxic protease and this formation is inhibited in the mutated non-toxic protease according to the present invention.

(3) FIG. **3** schematically shows the structures of a mutated non-toxic protease and a fusion non-toxic protease according to various embodiments of the present invention.

(4) FIG. **4** schematically shows the restriction enzyme positions used in recombinant vector cloning to produce various mutated non-toxic proteases and fusion mutated non-toxic proteases according to the present invention.

(5) FIG. **5** schematically shows the structure of a fusion non-cytotoxic protease into which (Gly).sub.4 capable of imparting structural flexibility to a fusion site between a mutated non-cytotoxic protease and a peptide (fusion partner) has been introduced and a cleavage amino acid sequence (LVPRGS) of the thrombin-like enzyme batroxobin, a protein cleavage enzyme, has been introduced so that it can cleave between the mutated non-toxic protease and the peptide (fusion partner) to form heterodimers linked by a disulfide bond.

(6) FIG. **6** schematically shows the structure of a fusion non-toxic protease for expression in the yeast *Pichia pastoris.*

(7) FIG. **7** shows the results of analyzing the expression patterns of mutated non-toxic proteases according to various embodiments of the present invention in *E. coli* by Coomassie blue staining.

(8) FIG. **8** shows the results of expressing a wild-type non-toxic protease in *E. coli* and analyzing the expression patterns thereof in the supernatant and precipitate by Coomassie blue staining and Western blotting.

(9) FIG. **9** shows the results of expressing a wild-type non-toxic protease in *E. coli* and analyzing the aggregation patterns of the wild-type non-toxic protease under non-reducing and reducing conditions by Coomassie blue staining and Western blotting.

(10) FIG. **10** shows the results of codon-optimizing the mutated non-toxic proteases according to various embodiments of the present invention so as to be suitable for expression in *E. coli,* and then analyzing the expression patterns thereof in *E. coli* by Coomassie blue staining.

(11) FIG. **11** shows the results of codon-optimizing the fusion non-toxic proteases according to various embodiments of the present invention so as to be suitable for expression in *E. coli,* and then analyzing the expression patterns thereof in *E. coli* by Coomassie blue staining.

(12) FIG. **12** shows the results of lysing *E. coli* cells expressing the mutated non-toxic proteases according to various embodiments of the present invention, separating the cell lysate into a soluble fraction and an insoluble fraction, and then analyzing the amount of a non-toxic protease in each fraction by Coomassie blue staining and Western blotting.

(13) FIG. **13** is a schematic diagram showing a production process according to the present invention.

(14) FIG. **14** shows the results of purifying the mutated non-toxic protease and fusion non-toxic protease according to various embodiments of the present invention, and then analyzing the non-toxic proteases by Coomassie blue staining and Western blotting.

(15) FIG. **15** shows the results of confirming the excellent refolding effects of the mutated non-toxic proteases and fusion non-toxic proteases according to various embodiments of the present invention.

(16) FIG. **16** shows the results of comparing the amounts of wild-type and mutated non-toxic proteases present in insoluble fractions (A) with the amounts of wild-type and mutant non-toxic

proteases recoverable by refolding (B).
(17) FIG. **17** shows the results of fusing an enzyme active substrate of a non-toxic protease with ANNEXIN V, expressing and purifying the fusion non-toxic protease in *E. coli*, and performing Coomassie blue staining, in order to analyze the cleavage activities of the mutated non-toxic protease and fusion non-toxic protease according to the present invention.
(18) FIG. **18** shows the results of analyzing the cleavage activities of the mutated non-toxic protease and fusion non-toxic protease according to the present invention by Coomassie blue staining using the SNAP25-ANNEXIN V fusion protein.
(19) FIG. **19** shows the results of expressing and purifying the fusion non-toxic protease in the recombinant yeast *Pichia pastoris* transformed with the fusion non-toxic protease according to the present invention, and analyzing the fusion non-toxic protease by Western blotting.
(20) FIG. **20** is a schematic diagram showing a process for fermenting and purifying a large amount of a non-toxic protease according to the present invention.
(21) FIG. **21** shows the results of performing SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a wild-type non-toxic protease.
(22) FIG. **22** shows the results of performing SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a mutated non-toxic protease (C430S) according to the present invention.
(23) FIG. **23** shows the results of SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a mutated non-toxic protease (C430A) according to the present invention.
(24) FIG. **24** shows the results of SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a mutated non-toxic protease (C430G) according to the present invention.
(25) FIG. **25** shows the results of SDS-PAGE electrophoresis and Coomassie staining after fermenting and purifying a large amount of a fusion non-toxic protease (C430G-BFGFRP) according to the present invention.
(26) FIG. **26** is a graph comparing the water-soluble protease formation rate and purification yield of each of the wild-type non-toxic protease and the mutant non-toxic protease according to the present invention.
DETAILED DESCRIPTION AND PREFERRED EMBODIMENTS OF THE INVENTION
(27) Unless otherwise defined, all technical and scientific terms used in the present specification have the same meanings as commonly understood by those skilled in the art to which the present disclosure pertains. In general, the nomenclature used in the present specification is well known and commonly used in the art.
(28) In the present invention, it has been found that, when a mutation at the amino acid cysteine at position 430 of a wild-type protease is induced in order to solve the problem associated with the decrease in productivity resulting from protein aggregation which frequently occurs during purification of the wild-type non-toxic protease in a process of producing the non-toxic protease using a recombinant microorganism, protein aggregation resulting from inclusion body formation is inhibited while the protease activity is maintained, and thus the non-toxic protease contained in an insoluble fraction can be recovered even by a simple refolding process alone, and even when various functional peptides are fused to the mutated non-toxic protease, the fusion mutated non-toxic protease can be produced with high productivity.
(29) Therefore, in one aspect, the present invention is directed to a mutated non-toxic protease in which the amino acid cysteine at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid other than cysteine.
(30) In the present invention, the amino acid cysteine at position 430 of the non-toxic protease is substituted with any amino acid other than cysteine so as not to form an inappropriate disulfide bond with another cysteine. The amino acid substituting for the cysteine at position 430 is not

limited, but the cysteine (Cys) may preferably be substituted with glycine (Gly), alanine (Ala) or serine (Ser).

(31) In another aspect, the present invention is directed to a fusion non-toxic proteinase in which the mutated non-toxic protease is fused with any one or more peptides selected from the group consisting of: i) a cell penetrating peptide; ii) a belt domain fragment peptide; and iii) a cell targeting peptide.

(32) In the present invention, the cell penetrating peptide may be represented by the amino acid sequence of SEQ ID NO: 3, the belt domain fragment peptide may be represented by the amino acid sequence of SEQ ID NO: 5, and the cell targeting peptide may be represented by the amino acid sequence of SEQ ID NO: 7, but the present invention is not limited thereto. Preferably, the fusion non-toxic protease is for expression in *E. coli*.

(33) In still another aspect, the present invention is directed to a fusion non-toxic proteinase in which the mutated non-toxic protease is fused with any one or more peptides selected from the group consisting of: i) a yeast secretion signal peptide; ii) a cell penetrating peptide; and ii) a cell targeting peptide. Preferably, the fusion non-toxic protease is for expression in yeast.

(34) In the present invention, the yeast secretion signal peptide may be represented by the amino acid sequence of SEQ ID NO: 46, the cell penetrating peptide may be represented by the amino acid sequence of SEQ ID NO: 3, and the cell targeting peptide may be represented by the amino acid sequence of SEQ ID NO: 7, but the present invention is not limited thereto.

(35) In the present invention, when the non-toxic protease constituting the fusion non-toxic protease is fused with another peptide at its N-terminus, the fusion may be performed after removal of the amino acid methionine at position 1.

(36) In the present invention, a tag peptide for purification may further be fused to the mutated non-toxic protease or the fusion non-toxic protease. The tag peptide for purification may be selected from the group consisting of glutathione-S-transferase (GST), C-myc tag, a chitin-binding domain, streptavidin binding protein (SBP), a cellulose-binding domain, a calmodulin-binding peptide, S-tag, Strep-tag II, FLA, protein A, protein G, histidine affinity tag (HAT), poly-His, thioredoxin, pelB leader, and maltose binding protein (MBP), but is not limited thereto. It is possible use any tag peptide for purification that is used in a process of expressing and purifying a protein in a microorganism in the art. It will be apparent to those skilled in the art that the peptide for purification may be cleaved according to a method known in the art after the completion of purification, and a mutated non-toxic protease or fusion non-toxic protease from which the tag peptide for purification has been detached may be used for its original purpose.

(37) In the present invention, any one of the peptides (i.e., a peptide for fusion or purification) may be fused directly or via a linker to the non-toxic protease, and the linker sequence comprises about 3 to 20 amino acids, more preferably about 3 to 10 amino acids. The linker sequence is preferably flexible so that the non-toxic protease is not maintained in an undesirable conformation. The linker sequence may be used, for example, to space the non-toxic protease moiety apart from the fused peptide. Specifically, when the non-toxic protease is fused with two or more peptides selected from the group consisting of a cell penetrating peptide, a belt domain fragment peptide, a cell targeting peptide, a yeast secretion signal peptide, and a tag peptide for purification, the peptide linker sequence may be appropriately placed as needed between the non-toxic protease and the peptide and/or between any two or more selected peptides to provide molecular flexibility. In order to provide flexibility, the linker preferably predominantly comprises amino acids having small side chains, such as glycine, alanine and serine. Preferably, at least about 80 or 90 percent of the linker sequence comprises a glycine, alanine or serine residue, in particular a glycine or serine residue. One suitable linker sequence may be a peptide linker, preferably represented by (Gly).sub.N (wherein N is an integer ranging from 3 to 10), but is not limited thereto.

(38) The present invention may further comprise a peptide sequence cleavable by a protease for cleavage between the C-terminus of the non-toxic protease and the N-terminus of a peptide selected

from the group consisting of a cell penetrating peptide, a belt domain fragment peptide, a cell targeting peptide, a yeast secretion signal peptide, and a tag peptide for purification.
(39) In the present invention, the cleavable peptide sequence may be LVPRGS, which is one of the cleavage sequences of the thrombin-like enzyme batroxobin, but is not limited thereto. In one embodiment, the cleavable peptide sequence is derived from the cleavage sequence of human fibrinogen alpha-chain, which is one of the cleavage sequences of the thrombin-like enzyme batroxobin, which is a protein cleaving enzyme. When the peptide fused with a non-toxic protease is cleaved by the cleavage sequence, the non-toxic protease may be linked to the cleaved peptide by a disulfide bond to form a heterodimer.
(40) However, any proteases that may cleave the peptide fused with the non-toxic protease include, without limitation, trypsin, pepsin, Lys-C endoproteinase, Lys-N endoproteinase, arginyl, endopeptidase, plasmin, omptin and clostridial proteases, as described in EP2524963. In one embodiment, the protease for cleavage is trypsin or Lys-C endoproteinase. In one embodiment, the protease is a protease that cleaves a non-toxic protease non-native (i.e. exogenous) cleavage site. Non-native proteases that may be utilized include, but are not limited to, enterokinase (DDDDK↓) (SEQ ID NO: 50), factor Xa (IEGR↓ (SEQ ID NO: 51)/IDGR↓) (SEQ ID NO: 52), TEV (Tobacco Etch Virus) (ENLYFQ↓G) (SEQ ID NO: 53), thrombin (LVPR↓GS) (SEQ ID NO: 54), and precleavage (LEVLFQ↓GP) (SEQ ID NO: 55), (↓ indicates a cleavage site). In the present invention, the cell penetrating peptide may be selected from the group consisting of Protein-derived Penetration (RQIKIWFQNRRMKWKK) (SEQ ID NO: 56), Tat peptide (GRKKRRQRRRPPQ) (SEQ ID NO: 57), pVEC (LLIILRRRIRKQAHAHSK) (SEQ ID NO: 58), chimeric transportan (GWTLNSAGYLLGKINLKALAALAKKIL) (SEQ ID NO: 59), MPG (GALFLGFLGAAGSTMGAWSQPKKKRKV) (SEQ ID NO: 60), Pep-1 (KETWWETWWTEWSQPKKKRKV) (SEQ ID NO: 61), synthetic Polyarginines ((R)n; 6<n<12), MAP (KLALKLALKALKAALKLA) (SEQ ID NO: 62), and R.sub.6W.sub.3 (RRWWRRWRR) (SEQ ID NO: 63), but is not limited thereto (see C. Bechara et al. FEBS Letter 587 (2013) 1693-1702).
(41) In one embodiment of the present invention, the cell targeting peptide may be NH2-Thr-Tyr-Arg-Ser-Arg-Lys-Tyr-(Thror Ser)-Ser-Trp-Tyr-COOH [TYRSRKY(S/T)SWY], which is a peptide derived from the amino acids at positions 105 to 115 of human basic fibroblast growth factor, but is not limited thereto. It will be obvious to those skilled in the art that any peptides used as targeting moieties for conventional non-toxic proteases, such as lectin wheat germ agglutinin, nerve growth factor (NGF), epidermal growth factor, an antibody fragment, or a growth hormone releasing hormone (GHRH) ligand (see Elena Fonfria et al., Toxins (Basel). 2018 July; 10(7): 278.) may be used in fusion with the non-toxic protease.
(42) In still another aspect, the present invention is directed to a mutant gene encoding the mutated non-toxic protease.
(43) In the present invention, the mutant gene may be represented by a nucleotide sequence selected from the group consisting of SEQ ID NOs: 30, 32 and 34.
(44) In yet another aspect, the present invention is directed to a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of i) a nucleic acid encoding a cell penetrating peptide; ii) a nucleic acid encoding a belt domain fragment peptide; and iii) a nucleic acid encoding a cell targeting peptide.
(45) In the present invention, the nucleic acid encoding the cell penetrating peptide may be represented by the nucleotide sequence of SEQ ID NO: 4, the nucleic acid encoding the belt domain fragment peptide may be represented by the nucleotide sequence of SEQ ID NO: 6, and the nucleic acid encoding the cell targeting peptide may be represented by the nucleotide sequence of SEQ ID NO: 8, but the present invention is not limited thereto.
(46) In another aspect, the present invention is directed to a gene construct comprising: the mutant gene; and any one or more nucleic acids selected from the group consisting of i) a nucleic acid

encoding a yeast secretion signal peptide; ii) a nucleic acid encoding a cell penetrating peptide; and iii) a nucleic acid encoding a cell targeting peptide.

(47) In the present invention, the nucleic acid encoding the yeast secretion signal peptide may be represented by the nucleotide sequence of SEQ ID NO: 47, the nucleic acid encoding the cell penetrating peptide may be represented by the nucleotide sequence of SEQ ID NO: 4, and the nucleic acid encoding the cell targeting peptide may be represented by the nucleotide sequence of SEQ ID NO: 8, but the present invention is not limited thereto.

(48) In the present invention, the mutant gene or gene construct may further comprise a nucleic acid encoding a tag peptide for purification. The tag peptide for purification may be selected from the group consisting of glutathione-S-transferase (GST), C-myc tag, a chitin-binding domain, streptavidin binding protein (SBP), a cellulose-binding domain, a calmodulin-binding peptide, S-tag, Strep-tag II, FLA, protein A, protein G, histidine affinity tag (HAT), poly-His, thioredoxin, pelB leader, and maltose binding protein (MBP), but is not limited thereto. It is possible to use any tag peptide for purification that is used in a process of expressing and purifying a protein in a microorganism in the art.

(49) In the present invention, the gene construct may further comprise a nucleic acid encoding the linker peptide that connects the peptide to the non-toxic protease, wherein the nucleic acid encoding the linker peptide may be represented by the nucleotide sequence of SEQ ID NO: 26.

(50) The present invention may further comprise a nucleic acid encoding a peptide sequence cleavable by a cleavage protease between the C-terminus of the non-toxic protease and the N-terminus of a peptide selected from the group consisting of a cell penetrating peptide, a belt domain fragment peptide, a cell targeting peptide, a yeast secretion signal peptide, and a tag peptide for purification. The nucleic acid encoding the cleavable peptide sequence may be represented by the nucleotide sequence of SEQ ID NO: 28, but is not limited thereto.

(51) The present invention encompasses fragments and variants of polypeptides (proteins) and genes (nucleic acids) provided as specific sequences as long as they maintain their original structural and/or functional characteristics. That is, these fragments and variants may have a sequence homology of at least 90%, more preferably at least 95%, at least 97% or at least 99% to the sequences provided in the present invention.

(52) In another aspect, the present invention is directed to a recombinant vector having the mutant gene introduced therein.

(53) In the present invention, as the recombinant vector, there may be used without limitation any vector known in the art that may be introduced into bacterial cells, yeast cells, mammalian cells, insect cells, plant cells, or amphibian cells and used for overexpression of recombinant proteins. Preferably, there may be used a recombinant vector that may be introduced into bacterial cells or yeast cells and used for overexpression of recombinant proteins. Preferably, the pET22b(+) (Novagen, Merk Millipore) vector may be used for expression in *E. coli* and the pPIC9 vector may be used for expression in yeast. However, it will be obvious to those skilled in the art that vectors usable in the present invention are not limited to these vectors, and vectors known in the art may be applied for the expression of the mutant non-toxic protease or fusion non-toxic protease of the present invention.

(54) In another aspect, the present invention is directed to a recombinant microorganism having the recombinant vector introduced therein.

(55) In the present invention, the recombinant microorganism may be bacteria or yeast, but is not limited thereto.

(56) In another aspect, the present invention is directed to a method for producing a mutated non-toxic protease or a fusion non-toxic protease, the method comprising steps of: (a) producing the mutated non-toxic protease or the fusion non-toxic protease by culturing the recombinant microorganism; (b) disrupting the recombinant microorganism; and (c) purifying the mutated non-toxic protease or the fusion non-toxic protease from the disrupted recombinant microorganism.

(57) In the present invention, step (a) may be performed in two steps: seed culture followed by main culture (i.e., culture for fermentation).
(58) In the present invention, in step (a), the recombinant microorganism may be cultured at 30 to 38° C., and then when the OD.sub.600 of the recombinant microorganism reaches 0.2 to 0.4, the culture temperature may be lowered to 16 to 20° C. and the recombinant microorganism may be further cultured. Preferably, in the main culture step in step (a), the recombinant microorganism may be cultured at 36 to 38° C., more preferably 36.5 to 37.5° C., and then when the OD.sub.600 of the recombinant microorganism reaches 0.25 to 0.35, the culture temperature may be 17 to 19° C. and the recombinant microorganism may be further cultured. In this case, protein expression may be induced by adding 0.4 to 0.6 mM IPTG to the culture medium when the OD.sub.600 reaches 0.5 to 0.7. Through this process, recovery of the mutated non-toxic protease or the fusion non-toxic protease according to the present invention from a soluble fraction is significantly increased. That is, in the initial stage, culture is performed at a high temperature in order to rapidly increase the amount of the recombinant microorganism, and then when the recombinant microorganism reaches a certain amount, the culture temperature of the recombinant microorganism is lowered to slow the metabolism of the recombinant microorganism. In addition, expression of a non-toxic protease is gradually induced by adding a small amount of IPTG, so that the refolding efficiency of the non-toxic protease is increased, and thus most of the non-toxic protease may be included in the soluble fraction. This is demonstrated by the results obtained in the present invention.
(59) The effect of increasing the expression level of the protease in the soluble fraction is particularly pronounced in the case of the mutated protease C430G derived in the present invention. This is believed to be because, in the case of a wild-type mutant protease, a disulfide bond between cysteines is induced, and in the case of C430S and C430A, which are other mutated proteases, some non-specific disulfide bonds are induced.
(60) In the present invention, step (c) may comprise separating the disrupted recombinant microorganism into a soluble fraction and an insoluble fraction by centrifugation, and independently purifying the mutated non-toxic protease or the fusion non-toxic protease from each fraction.
(61) In the present invention, step (c) may comprise separating the disrupted recombinant microorganism into a soluble fraction and an insoluble fraction by centrifugation, and purifying the mutated non-toxic protease or the fusion non-toxic protease from the soluble fraction.
(62) In the present invention, suitable medium and culture conditions that are used to culture the recombinant microorganism may be those known in the art. For example, the recombinant microorganism may be inoculated and cultured in a suitable medium for transformed *E. coli* or yeast, and then expression of the protein may be induced under suitable conditions. For example, expression of the protein in *E. coli* may be induced by supplying isopropyl-β-D-thiogalactoside, and expression of the protein in isopropyl-β or methanol-assimilating yeast may be induced by supplying methyl alcohol. After completion of the step of inducing protein expression by culture, the culture or culture medium may be recovered, and “pure recombinant protein” may be recovered therefrom. As used herein, the term “pure recombinant protein” means that the recombinant protein does not contain 5% or more, preferably 1% or more of any other proteins derived from the host cell, other than the recombinant protein of the present invention and a protein derived from the DNA sequence encoding the same.
(63) Purification of the recombinant protein expressed in the transformed microorganism may be performed by various methods known in the art. Usually, the cell lysate or culture may be centrifuged to remove cell debris, culture impurities, etc., and then subjected to precipitation, for example, salting out (ammonium sulfate precipitation and sodium phosphate precipitation), solvent precipitation (protein fraction precipitation using acetone, ethanol, isopropyl alcohol, etc.) and the like, and subjected to dialysis, electrophoresis, and various types of column chromatography. As

the types of chromatography, techniques such as ion exchange chromatography, gel-filtration chromatography, HPLC, reverse phase HPLC, adsorption chromatography, affinity column chromatography and ultrafiltration may be used alone or in combination.

(64) In the present invention, the term “non-toxic protease” is meant to include a peptide that has significantly reduced toxicity due to removal of the heavy chain domain from the natural botulinum toxin type A and has the natural botulinum toxin type A light-chain domain having protease activity, as well as a variant of the peptide.

(65) In another aspect, the present invention is directed to a pharmaceutical composition for treating Dystonia containing the mutated or fusion non-toxic protease as an active ingredient.

(66) In the present invention, the pharmaceutical composition may be for transdermal administration, but is not limited thereto.

(67) In another aspect, the present invention is directed to a method for alleviating or treating symptoms of Dystonia, the method comprising a step of administering the mutated or fusion non-toxic protease to a subject in need of alleviation or treatment of symptoms of Dystonia.

(68) In another aspect, the present invention is directed to the use of the mutated or fusion non-toxic protease for the manufacture of a medicament for use in the alleviation or treatment of symptoms of Dystonia.

(69) In another aspect, the present invention is directed to the use of the above-described mutated or fusion non-toxic protease for use in a method for alleviating or treating symptoms of Dystonia.

(70) In the present invention, the muscular dystonia may be selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis, but is not limited thereto.

(71) In another aspect, the present invention is directed to a hyaluronic acid microneedle patch containing the mutated or fusion non-toxic protease.

(72) In the present invention, the hyaluronic acid microneedle patch may be used for the treatment or alleviation of a symptom selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis, but is not limited thereto.

(73) As used herein, the term “treating” means reversing, alleviating, inhibiting the progression of, or preventing a disorder or disease to which the term applies, or one or more symptoms of the disorder or disease, unless otherwise stated. As used herein, the term “treatment” refers to the act of treating when ‘treating’ is defined as above. As used herein, the term “treatment” refers to the act of treating when the term “treating” is defined as above. Accordingly, treatment or therapy of the disease in a mammal may include one or more of the following: (1) inhibiting the development of the disease; (2) preventing the spread of the disease; (3) relieving the disease; (4) preventing the recurrence of the disease; and (5) palliating symptoms of the disease.

(74) As used herein, the term “effective amount” means an amount that is high enough to deliver the desired benefit, but low enough to avoid serious side effects within the scope of medical judgment. The amount of the non-toxic protease that is administered into the body by the composition of the present invention may be appropriately adjusted in consideration of the route of administration and the for administration.

(75) The composition of the present invention may be administered to the subject once daily, once every several days, once every several months, or once or more every several years. Unit dosage means physically discrete units suitable for unit administration to human subjects and other mammals, each unit comprising a suitable pharmaceutical carrier and a predetermined amount of the protease of the present invention that exhibits a therapeutic effect. However, the dosage may vary depending on the severity of the patient's disease and the active ingredient and auxiliary active ingredient used. In addition, the total daily dosage may be divided into several times and continuously administered as needed. Accordingly, the above dosage range is not intended to limit

the scope of the present invention in any way.

(76) As used herein, the term “pharmaceutically acceptable” refers to a composition which is physiologically acceptable and, when administered to humans, does not cause allergic reactions such as gastrointestinal disorders and dizziness, or similar reactions.

(77) The pharmaceutical composition of the present invention may be formulated using a method known in the art so as to provide quick, sustained or delayed release of the active ingredient after administration to mammals. The dosage forms may be powders, granules, tablets, emulsions, syrups, aerosols, soft or hard gelatin capsules, sterile injection solutions, sterile powders, or patches. In addition, the composition for preventing or treating a disease according to the present invention may be administered through several routes, including oral, transdermal, subcutaneous, intravenous and intramuscular routes. The dosage of the active ingredient may be suitably selected depending on various factors such as the route of administration, and patient's age, sex, weight and severity, and the active ingredient may be administered in combination with a known compound having the effect of preventing, alleviating or treating the symptoms of the disease.

(78) Hereinafter, the present invention will be described in more detail with reference to examples. It will be obvious to those skilled in the art that these examples serve only to illustrate the present invention, and the scope of the present invention is not limited by these examples.

Example 1. Construction of Vectors for Expression of Mutated Non-Toxic Proteases and Fusion Non-Toxic Proteases

(79) As shown in FIG. **3**, expression vectors for producing a mutated non-toxic protease and a fusion non-toxic protease were constructed. To this end, the following nucleotide sequences were synthesized: a nucleotide sequence (SEQ ID NO: 2) encoding a wild-type non-toxic protease (BTX-LC) represented by the amino acid sequence of SEQ ID NO: 1; a nucleotide sequence (SEQ ID NO: 4) encoding a cell penetrating peptide (PEP1); a nucleotide sequence (SEQ ID NO: 6) encoding a belt domain peptide fragment (belt'); and a nucleotide sequence (SEQ ID NO: 8) encoding a targeting peptide.

(80) Based on the synthesized nucleotide sequences, BTX-LC and PEP1-(ΔM)BTX-L-His were cloned by restriction enzymes (Ndel and XhoI) into the multiple cloning site (MCS) downstream of the T7 promoter-lac operator of the expression vector pET22b for *E. coli*. Thereafter, using the restriction enzymes EcoRI and XhoI, recombinant vectors capable of expressing various non-toxic proteases shown in Table 1 below, including the following non-toxic proteases, were constructed: PEP1-(Δ)BTX-L-His (*E. coli* codon optimization; theoretical pI/Mw: 8.66/55003.59 Da) capable of improving expression efficiency in *E. coli*; a mutated non-toxic protease in which the amino acid cysteine (Cys) at position 430 of the wind-type non-toxic protease is substituted with each of glycine (Gly), alanine (Ala) and serine (Ser); and a PEP1-(ΔM)BTX-L-BFGFRP-His (theoretical pI/Mw: 8.95/56436.13) fusion non-toxic protease. Table 2 below shows the sequences (gene cassettes for cloning of mutated non-toxic proteases) used in cloning for a point mutation at the amino acid at position 430 of the wild-type protease.

(81) TABLE-US-00001 TABLE 1 Amino acid sequence Nucleotide sequence BTX-LC MQFVNKQFNYKDPVNGVDIAYIKIPNVG 5′- QMQPVKAFKIHNKIWVIPERDTFTNPEEG
atgcaatttgttaataaacaatttaattataaagatcctgtaaatggtgttgatat
DLNPPPEAKQVPVSYYDSTYLSTDNEKDN
tgcttatataaaaattccaaatgtaggacaaatgcaaccagtaaaagctttta
YLKGVTKLFERIYSTDLGRMLLTSIVRGIP
aaattcataataaaatatgggttattccagaaagagatacatttacaaatcct
FWGGSTIDTELKVIDTNCINVIQPDGSYRS
gaagaaggagatttaaatccaccaccagaagcaaaacaagttccagtttca
EELNLVIIGPSADIIQFECKSFGHEVLNLTR
tattatgattcaacatatttaagtacagataatgaaaaagataattatttaaagg
NGYGSTQYIRFSPDFTFGFEESLEVDTNPL

gagttacaaaattatttgagagaatttattcaactgatcttggaagaatgttgtt
LGAGKFATDPAVTLAHELIHAGHRLYGIA
aacatcaatagtaaggggaataccattttggggtggaagtacaatagatac
INPNRVFKVNTNAYYEMSGLEVSFEELRT
agaattaaaagttattgatactaattgtattaatgtgatacaaccagatggtag
FGGHDAKFIDSLQENEFRLYYYNKFKDIA
ttatagatcagaagaacttaatctagtaataataggaccctcagctgatattat
STLNKAKSIVGTTASLQYMKNVFKEKYLL
acagtttgaatgtaaaagctttggacatgaagttttgaatcttacgcgaaatg
SEDTSGKFSVDKLKFDKLYKMLTEIYTED
gttatggctctactcaatacattagatttagcccagattttacatttggttttgag
NFVKFFKVLNRKTYLNFDKAVFKINIVPK
gagtcacttgaagttgatacaaatcctcttttaggtgcaggcaaatttgctac
VNYTIYDGFNLRNTNLAANFNGQNTEINN
agatccagcagtaacattagcacatgaacttatacatgctggacatagattat
MNFTKLKNFTGLFEFYKLLCVRGIITSKTK
atggaatagcaattaatccaaatagggtttttaaagtaaatactaatgcctatt SLDKGYNK (SEQ ID NO: 1)
atgaaatgagtgggttagaagtaagctttgaggaacttagaacatttgggg
gacatgatgcaaagtttatagatagtttacaggaaaacgaatttcgtctatatt
attataataagtttaaagatatagcaagtacacttaataaagctaaatcaatag
taggtactactgcttcattacagtatatgaaaaatgtttttaaagagaaatatct
cctatctgaagatacatctggaaaattttcggtagataaattaaaatttgataa
gttatacaaaatgttaacagagatttacacagaggataattttgttaagtttttt
aaagtacttaacagaaaaacatatttgaattttgataaagccgtatttaagata
aatatagtacctaaggtaaattacacaatatatgatggatttaatttaagaaat
acaaatttagcagcaaactttaatggtcaaaatacagaaattaataatatgaa
ttttactaaactaaaaaattttactggattgtttgaattttataagttgctatgtgt
aagagggataataacttctaaaactaaatcattagataaaggatacaataag- 3′ (SEQ ID NO: 2) PEP1
MKETWWETWWTEWSQPKKK<u>R</u>KV (SEQ 5′- ID NO: 3)
atgaaggaaacttggtgggaaacttggtggactgaatggtctcaaccaaag aagaag<u>aga</u>aaggtt-3′ (SEQ ID NO: 4)
Belt ALNDLC (SEQ ID NO: 5) <u>Gcgctgaacgatctgtgc</u> (SEQ ID NO: 6) BFGFRP
TYRSRKYXSWY (SEQ ID NO: 7) 5′-ACCTATCGCAGCCGCAAATATASC (X stands for S or T.) AGCTGGTAT-3′ (SEQ ID NO: 8) (ASC stands for <u>AGC or ACC</u>.) PEP1- MKETWWETWWTEWSQPKKK<u>R</u>KVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
atgaaggaaacttggtgggaaacttggtggactgaatggtctcaaccaaag BTX-L-His
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagagaaaggttcaatttgttaataaacaatttaattataaagatcctgta
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
aatggtgttgatattgcttatataaaaattccaaatgtaggacaaatgcaacc
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
agtaaaagcttttaaaattcataataaaatatgggttattccagaaagagata
TIDTELKVIDTNCINVIQPDGSYRSEELNL
catttacaaatcctgaagaaggagatttaaatccaccaccagaagcaaaac
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aagttccagtttcatattatgattcaacatatttaagtacagataatgaaaaag
TQYIRFSPDFTFGFEESLEVDTNPLLGAGK
ataattatttaaagggagttacaaaattatttgagagaatttattcaactgatctt FATDPAVTLAHELIHAGHRLYGIAINPNR
ggaagaatgttgttaacatcaatagtaaggggaataccattttggggtggaa
VFKVNTNAYYEMSGLEVSFEELRTFGGH

gtacaatagatacagaattaaaagttattgatactaattgtattaatgtgatac
DAKFIDSLQENEFRLYYYNKFKDIASTLN
aaccagatggtagttatagatcagaagaacttaatctagtaataataggacc
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
ctcagctgatattatacagtttgaatgtaaaagctttggacatgaagttttgaa
SGKESVDKLKEDKLYKMLTEIYTEDNEVK
tcttacgcgaaatggttatggctctactcaatacattagatttagcccagatttt
FEKVLNRKTYLNEDKAVFKINIVPKVNYTI
acatttggttttgaggagtcacttgaagttgatacaaatcctcttttaggtgca
YDGENLRNTNLAANENGQNTEINNMNFT
ggcaaatttgctacagatccagcagtaacattagcacatgaacttatacatg
KLKNFTGLFEFYKLLCVRGIITSKTKSLDK
ctggacatagattatatggaatagcaattaatccaaatagggtttttaaagta GYNKHHHHHH (SEQ ID NO: 9)
aatactaatgcctattatgaaatgagtgggttagaagtaagctttgaggaact
tagaacatttgggggacatgatgcaaagtttatagatagtttacaggaaaac
gaatttcgtctatattattataataagtttaaagatatagcaagtacacttaata
aagctaaatcaatagtaggtactactgcttcattacagtatatgaaaaatgttt
ttaaagagaaatatctcctatctgaagatacatctggaaaattttcggtagata
aattaaaatttgataagttatacaaaatgttaacagagatttacacagaggat
aattttgttaagtttttaaagtacttaacagaaaaacatatttgaattttgataaa
gccgtatttaagataaatatagtacctaaggtaaattacacaatatatgatgg
atttaatttaagaaatacaaatttagcagcaaactttaatggtcaaaatacaga
aattaataatatgaattttactaaactaaaaaattttactggattgtttgaatttta
taagttgctatgtgtaagagggataataacttctaaaactaaatcattagataa aggatacaataagcatcaccatcaccatcactaa-3′
(SEQ ID NO: 10) PEP1- MKETWWETWWTEWSQPKKKRKVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L-
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc His
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc *(E. coli*
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc codon
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca opti-
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga mization)
TQYIRFSPDFTFGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta
DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtttgaatgtaaaagctttggtcat
SGKFSVDKLKFDKLYKMLTEIYTEDNFVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FFKVLNRKTYLNFDKAVFKINIVPKVNYTI
tagcccagattttacctttggttagaggagagcctggaagttgataccaatc

YDGFNLRNTNLAANFNGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**C**VRGHTSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKHHHHHH
atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa (SEQ ID NO: 11)
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc
tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagtttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttgaattctataagctgctg**TGT**gtacgcggtat
catcaccagcaaaaccaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataactcgag-3′ (SEQ ID NO: 12) PEP1- MKETWWETWWTEWSQPKKKRKVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc (C430G)-
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc His
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga
TQYIRFSPDFTFGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta
DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtagaatgtaaaagctttggtcat
SGKFSVDKLKFDKLYKMLTEIYTEDNFVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FFKVLNRKTYLNFDKAVFKINIVPKVNYTI
tagcccagatatacctaggattgaggagagcctggaagttgataccaatc
YDGFNLRNTNLAANFNGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**G**VRGIITSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKHHHHHH (SEQ ID NO: 13)
atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc

tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagtttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttgaattctataagctgctg**GGC**gtacgcggtat
catcaccagcaaaaccaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataactcgag-3′ (SEQ ID NO: 14) PEP1- MKETWWETWWTEWSQPKKKRKVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc (C430A)-
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc His
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga
TQYIRFSPDFTEGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta
DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtagaatgtaaaagctttggtcat
SGKESVDKLKEDKLYKMLTEIYTEDNEVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FEKVLNRKTYLNEDKAVFKINIVPKVNYTI
tagcccagatatacctaggattgaggagagcctggaagttgataccaatc
YDGENLRNTNLAANENGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**A**VRGHTSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKHHHHHH (SEQ ID NO: 15)
atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc
tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagtttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttgaattctataagctgctg**GCG**gtacgcggtat
catcaccagcaaaaccaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataactcgag-3′ (SEQ ID

NO: 16) PEP1- MKETWWETWWTEWSQPKKKRKVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc (C430S)-
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc His
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga
TQYIRFSPDFTEGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta
DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtagaatgtaaaagctttggtcat
SGKESVDKLKEDKLYKMLTEIYTEDNEVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FEKVLNRKTYLNEDKAVFKINIVPKVNYTI
tagcccagattttacctaggattgaggagagcctggaagttgataccaatc
YDGENLRNTNLAANENGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**S**VRGIITSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKHHHHHH (SEQ ID NO: 17)
atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc
tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagtttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttgaattctataagctgctg**AGC**gtacgcggtat
catcaccagcaaaaccaaaagcctggataaaggctacaataagcatcacc atcaccatcactaataactcgag-3′ (SEQ ID
NO: 18) PEP1- MKETWWETWWTEWSQPKKKRKVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc (C430G)-
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc BFGFRP-
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS

aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc His
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga
TQYIRFSPDFTEGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta
DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtttgaatgtaaaagattggtcat
SGKFSVDKLKFDKLYKMLTEIYTEDNFVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FFKVLNRKTYLNFDKAVFKINIVPKVNYTI
tagcccagattttacctttggattgaggagagcctggaagttgataccaatc
YDGFNLRNTNLAANFNGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**G**VRGIITSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKTYRSRKYXSWYHHHHHH
(SEQ ID atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa NO: 19)
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga (X stands for S or T.)
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc
tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagtttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttg<u>aattc</u>tataagctgctg**GGCg**TACGCG
GTATCATCACCAGCAAAACCAAAAGCCTGGA
TAAAGGCTACAATAAGACCTATCGCAGCCGC
AAATAT<u>**ASC**</u>AGCTGGTATCATCACCATCACCA TCACTAATAA<u>CTCGAG</u>-3′ (SEQ ID NO: 20) (ASC stands for <u>AGC or ACC</u>.) PEP1-
MKETWWETWWTEWSQPKKK<u>R</u>KVQFVN 5′- (ΔM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc (C430A)-
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc BFGFRP-
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc His
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga

TQYIRFSPDFTFGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta
DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtttgaatgtaaaagattggtcat
SGKFSVDKLKFDKLYKMLTEIYTEDNFVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FFKVLNRKTYLNFDKAVFKINIVPKVNYTI
tagcccagattttacctttggattgaggagagcctggaagttgataccaatc
YDGFNLRNTNLAANFNGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**A**VRGIITSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKTYRSRKYXSWYHHHHHH
(SEQ ID atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa NO: 21)
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga (X stands for S or T.)
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc
tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagtttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttgaattctataagctgctg**GCG**gTACGCG
GTATCATCACCAGCAAAACCAAAAGCCTGGA
TAAAGGCTACAATAAGACCTATCGCAGCCGC
AAATAT**ASC**AGCTGGTATCATCACCATCACCA TCACTAATAACTCGAG-3′ (SEQ ID NO: 22) (ASC stands for AGC or ACC.) PEP1-
MKETWWETWWTEWSQPKKKRKVQFVN 5′- (AM)
KQFNYKDPVNGVDIAYIKIPNVGQMQPV
catatgaaggaaacttggtgggaaacttggtggaccgaatggtctcaacca BTX-L
KAFKIHNKIWVIPERDTFTNPEEGDLNPPP
aagaagaagcgcaaggttcaatttgttaataaacaatttaattataaagatcc (C430S)-
EAKQVPVSYYDSTYLSTDNEKDNYLKGV
agtaaatggtgtcgacattgcttatatcaaaattccaaatgtaggccaaatgc BFGFRP-
TKLFERIYSTDLGRMLLTSIVRGIPFWGGS
aaccagtaaaagcttttaaaattcataataaaatctgggttattccagaacgc His
TIDTELKVIDTNCINVIQPDGSYRSEELNL
gatacctttaccaatccggaagaaggtgatctgaatccaccaccagaagca
VIIGPSADIIQFECKSFGHEVLNLTRNGYGS
aaacaagttccagttagctattatgatagcacctatctgagcaccgataatga
TQYIRFSPDFTFGFEESLEVDTNPLLGAGK
aaaagataattatctgaagggcgttaccaaactgtttgagcgcatttatagca FATDPAVTLAHELIHAGHRLYGIAINPNR
ctgatctgggtcgcatgctgctgaccagcatcgtacgcggtatcccattttg
VFKVNTNAYYEMSGLEVSFEELRTFGGH
gggtggtagcaccatcgataccgaactgaaagttattgatactaattgtatta

DAKFIDSLQENEFRLYYYNKFKDIASTLN
atgtgatccaaccagatggtagctatcgcagcgaagaactgaatctggtaa
KAKSIVGTTASLQYMKNVFKEKYLLSEDT
tcatcggtccgagcgctgatattatccagtttgaatgtaaaagattggtcat
SGKFSVDKLKFDKLYKMLTEIYTEDNFVK
gaagttctgaatctgacccgtaatggttatggcagcacccaatacattcgctt
FFKVLNRKTYLNFDKAVFKINIVPKVNYTI
tagcccagattttacctttggattgaggagagcctggaagttgataccaatc
YDGFNLRNTNLAANFNGQNTEINNMNFT
cgctgctgggtgcaggcaaatttgctaccgatccagcagtaaccctggca
KLKNFTGLFEFYKLL**S**VRGIITSKTKSLDK
catgaactgatacatgctggccatcgcctgtatggtatcgcaattaatccaa GYNKTYRSRKYXSWYHHHHHH
(SEQ ID atcgcgtttttaaagtaaataccaatgcctattatgaaatgagcggtctggaa NO: 23)
gtaagctttgaggaactgcgcacctttggtggtcatgatgcaaagtttatcga (X stands for S or T.)
tagcctgcaggaaaacgaatttcgtctgtattattataataagtttaaagatat
cgcaagcaccctgaataaagctaaaagcatcgtaggtaccaccgctagcc
tgcagtatatgaaaaatgtttttaaagagaaatatctgctgtctgaagatacct
ctggcaaatttagcgtagataaactgaaatttgataagctgtacaaaatgctg
accgagatttacaccgaggataattttgttaagttttttaaagtactgaaccgc
aaaacctatctgaattttgataaagccgtatttaagatcaatatcgtaccgaa
ggtaaattacaccatctatgatggttttaatctgcgcaataccaatctggcag
caaactttaatggtcaaaataccgaaattaataatatgaattttaccaaactga
aaaattttaccggtctgtttgaattctataagctgctg**AGCg**TACGCG
GTATCATCACCAGCAAAACCAAAAGCCTGGA
TAAAGGCTACAATAAGACCTATCGCAGCCGC
AAATAT**ASC**AGCTGGTATCATCACCATCACCA TCACTAATAACTCGAG-3′ (SEQ ID NO: 24) (ASC stands for AGC or ACC.) Linker GGGG (SEQ ID NO: 25)
GGTGGTGGTGGT (SEQ ID NO: 26) Cleavage LVPRGS (SEQ ID NO: 27)
CTGGTACCACGCGGTAGC (SEQ ID NO: 28) Peptide BTX-L
MQFVNKQFNYKDPVNGVDIAYIKIPNVG 5′- (C430G)
QMQPVKAFKIHNKIWVIPERDTFTNPEEG
atgcaatttgttaataaacaatttaattataaagatccagtaaatggtgtcgac
DLNPPPEAKQVPVSYYDSTYLSTDNEKDN
attgcttatatcaaaattccaaatgtaggccaaatgcaaccagtaaaagcttt
YLKGVTKLFERIYSTDLGRMLLTSIVRGIP
taaaattcataataaaatctgggttattccagaacgcgatacctttaccaatcc
FWGGSTIDTELKVIDTNCINVIQPDGSYRS
ggaagaaggtgatctgaatccaccaccagaagcaaaacaagttccagtta
EELNLVIIGPSADIIQFECKSFGHEVLNLTR
gctattatgatagcacctatctgagcaccgataatgaaaaagataattatctg
NGYGSTQYIRFSPDFTEGFEESLEVDTNPL
aagggcgttaccaaactgtttgagcgcatttatagcactgatctgggtcgca
LGAGKFATDPAVTLAHELIHAGHRLYGIA
tgctgctgaccagcatcgtacgcggtatcccattttggggtggtagcaccat
INPNRVFKVNTNAYYEMSGLEVSFEELRT
cgataccgaactgaaagttattgatactaattgtattaatgtgatccaaccag
FGGHDAKFIDSLQENEFRLYYYNKFKDIA
atggtagctatcgcagcgaagaactgaatctggtaatcatcggtccgagcg
STLNKAKSIVGTTASLQYMKNVFKEKYLL
ctgatattatccagtttgaatgtaaaagctttggtcatgaagttctgaatctga

SEDTSGKESVDKLKEDKLYKMLTEIYTED
cccgtaatggttatggcagcacccaatacattcgctttagcccagattttacc
NFVKFFKVLNRKTYLNFDKAVFKINIVPK
tttggttttgaggagagcctggaagttgataccaatccgctgctgggtgcag
VNYTIYDGFNLRNTNLAANFNGQNTEINN
gcaaatttgctaccgatccagcagtaaccctggcacatgaactgatacatg
MNFTKLKNFTGLFEFYKLL**G**VRGIITSKT
ctggccatcgcctgtatggtatcgcaattaatccaaatcgcgtttttaaagta KSLDKGYNK
aataccaatgcctattatgaaatgagcggtctggaagtaagctttgaggaac (SEQ ID NO: 29)
tgcgcacctttggtggtcatgatgcaaagtttatcgatagcctgcaggaaaa
cgaatttcgtctgtattattataataagtttaaagatatcgcaagcaccctgaa
taaagctaaaagcatcgtaggtaccaccgctagcctgcagtatatgaaaaa
tgtttttaaagagaaatatctgctgtctgaagatacctctggcaaatttagcgt
agataaactgaaatttgataagctgtacaaaatgctgaccgagatttacacc
gaggataattttgttaagtttttaaagtactgaaccgcaaaacctatctgaatt
ttgataaagccgtatttaagatcaatatcgtaccgaaggtaaattacaccatc
tatgatggattaatctgcgcaataccaatctggcagcaaactttaatggtca
aaataccgaaattaataatatgaattttaccaaactgaaaaattttaccggtct
gtttg<u>aattc</u>tataagctgctg<u>**GGC**</u>gtacgcggtatcatcaccagcaaaa ccaaaagcctggataaaggctacaataag-3′ (SEQ ID NO: 30) BTX-L MQFVNKQFNYKDPVNGVDIAYIKIPNVG 5′- (C430A)
QMQPVKAFKIHNKIWVIPERDTFTNPEEG
atgcaatttgttaataaacaatttaattataaagatccagtaaatggtgtcgac
DLNPPPEAKQVPVSYYDSTYLSTDNEKDN
attgcttatatcaaaattccaaatgtaggccaaatgcaaccagtaaaagcttt
YLKGVTKLFERIYSTDLGRMLLTSIVRGIP
taaaattcataataaaatctgggttattccagaacgcgataccataccaatcc
FWGGSTIDTELKVIDTNCINVIQPDGSYRS
ggaagaaggtgatctgaatccaccaccagaagcaaaacaagttccagtta
EELNLVIIGPSADIIQFECKSFGHEVLNLTR
gctattatgatagcacctatctgagcaccgataatgaaaaagataattatctg
NGYGSTQYIRFSPDFTFGFEESLEVDTNPL
aagggcgttaccaaactgtttgagcgcatttatagcactgatctgggtcgca
LGAGKFATDPAVTLAHELIHAGHRLYGIA
tgctgctgaccagcatcgtacgcggtatcccattttggggtggtagcaccat
INPNRVFKVNTNAYYEMSGLEVSFEELRT
cgataccgaactgaaagttattgatactaattgtattaatgtgatccaaccag
FGGHDAKFIDSLQENEFRLYYYNKFKDIA
atggtagctatcgcagcgaagaactgaatctggtaatcatcggtccgagcg
STLNKAKSIVGTTASLQYMKNVFKEKYLL
ctgatattatccagtttgaatgtaaaagctttggtcatgaagttctgaatctga
SEDTSGKFSVDKLKFDKLYKMLTEIYTED
cccgtaatggttatggcagcacccaatacattcgctttagcccagattttacc
NFVKFFKVLNRKTYLNFDKAVFKINIVPK
tttggttttgaggagagcctggaagttgataccaatccgctgctgggtgcag
VNYTIYDGFNLRNTNLAANFNGQNTEINN
gcaaatttgctaccgatccagcagtaaccctggcacatgaactgatacatg
MNFTKLKNFTGLFEFYKLL**A**VRGIITSKTK
ctggccatcgcctgtatggtatcgcaattaatccaaatcgcgtttttaaagta SLDKGYNK
aataccaatgcctattatgaaatgagcggtctggaagtaagctttgaggaac (SEQ ID NO: 31)
tgcgcacctttggtggtcatgatgcaaagtttatcgatagcctgcaggaaaa

cgaatttcgtctgtattattataataagtttaaagatatcgcaagcaccctgaa
taaagctaaaagcatcgtaggtaccaccgctagcctgcagtatatgaaaaa
tgtttttaaagagaaatatctgctgtctgaagatacctctggcaaatttagcgt
agataaactgaaatttgataagctgtacaaaatgctgaccgagatttacacc
gaggataattttgttaagttttttaaagtactgaaccgcaaaacctatctgaatt
ttgataaagccgtatttaagatcaatatcgtaccgaaggtaaattacaccatc
tatgatggattaatctgcgcaataccaatctggcagcaaactttaatggtca
aaataccgaaattaataatatgaattttaccaaactgaaaaattttaccggtct
gtttg<u>aattc</u>tataagctgctg<u>**GCG**</u>gtacgcggtatcatcaccagcaaaa ccaaaagcctggataaaggctacaataag-3′ (SEQ ID NO: 32) BTX-L MQFVNKQFNYKDPVNGVDIAYIKIPNVG 5′- (C430S)
QMQPVKAFKIHNKIWVIPERDTFTNPEEG
atgcaatttgttaataaacaatttaattataaagatccagtaaatggtgtcgac
DLNPPPEAKQVPVSYYDSTYLSTDNEKDN
attgcttatatcaaaattccaaatgtaggccaaatgcaaccagtaaaagcttt
YLKGVTKLFERIYSTDLGRMLLTSIVRGIP
taaaattcataataaaatctgggttattccagaacgcgataccataccaatcc
FWGGSTIDTELKVIDTNCINVIQPDGSYRS
ggaagaaggtgatctgaatccaccaccagaagcaaaacaagttccagtta
EELNLVIIGPSADIIQFECKSFGHEVLNLTR
gctattatgatagcacctatctgagcaccgataatgaaaaagataattatctg
NGYGSTQYIRFSPDFTFGFEESLEVDTNPL
aagggcgttaccaaactgtttgagcgcatttatagcactgatctgggtcgca
LGAGKFATDPAVTLAHELIHAGHRLYGIA
tgctgctgaccagcatcgtacgcggtatcccattttggggtggtagcaccat
INPNRVFKVNTNAYYEMSGLEVSFEELRT
cgataccgaactgaaagttattgatactaattgtattaatgtgatccaaccag
FGGHDAKFIDSLQENEFRLYYYNKFKDIA
atggtagctatcgcagcgaagaactgaatctggtaatcatcggtccgagcg
STLNKAKSIVGTTASLQYMKNVFKEKYLL
ctgatattatccagtttgaatgtaaaagctttggtcatgaagttctgaatctga
SEDTSGKFSVDKLKFDKLYKMLTEIYTED
cccgtaatggttatggcagcacccaatacattcgctttagcccagattttacc
NFVKFFKVLNRKTYLNFDKAVFKINIVPK
tttggttttgaggagagcctggaagttgataccaatccgctgctgggtgcag
VNYTIYDGFNLRNTNLAANFNGQNTEINN
gcaaatttgctaccgatccagcagtaaccctggcacatgaactgatacatg
MNFTKLKNFTGLFEFYKLL**S**VRGIITSKTK
ctggccatcgcctgtatggtatcgcaattaatccaaatcgcgtttttaaagta SLDKGYNK
aataccaatgcctattatgaaatgagcggtctggaagtaagctttgaggaac (SEQ ID NO: 33)
tgcgcacctttggtggtcatgatgcaaagtttatcgatagcctgcaggaaaa
cgaatttcgtctgtattattataataagtttaaagatatcgcaagcaccctgaa
taaagctaaaagcatcgtaggtaccaccgctagcctgcagtatatgaaaaa
tgtttttaaagagaaatatctgctgtctgaagatacctctggcaaatttagcgt
agataaactgaaatttgataagctgtacaaaatgctgaccgagatttacacc
gaggataattttgttaagttttttaaagtactgaaccgcaaaacctatctgaatt
ttgataaagccgtatttaagatcaatatcgtaccgaaggtaaattacaccatc
tatgatggattaatctgcgcaataccaatctggcagcaaactttaatggtca
aaataccgaaattaataatatgaattttaccaaactgaaaaattttaccggtct
gtttg<u>aattc</u>tataagctgctg<u>**AGC**</u>gtacgcggtatcatcaccagcaaaa ccaaaagcctggataaaggctacaataag-3′ (SEQ ID NO: 34)

(82) TABLE-US-00002 TABLE 2 Mutation Sequence C430G GAATTCTATAAGCTGCTGGGCGTACGCGGTATCATCACCAGCAAAACCAAAAG CCTGGATAAAGGCTACAATAAGCATCACCATCACCATCACTAATAACTCGAG (SEQ ID NO: 35) C430A GAATTCTATAAGCTGCTGGCGGTACGCCGGTATCATCACCAGCAAAACCAAAAG CCTGGATAAAGGCTACAATAAGCATCACCATCACCATCACTAATAACTCGAG (SEQ ID NO: 36) C430S GAATTCTATAAGCTGCTGAGCGTACGCGGTATCATCACCAGCAAAACCAAAAG CCTGGATAAAGGCTACAATAAGCATCACCATCACCATCACTAATAACTCGAG (SEQ ID NO: 37) C430G-GAATTCTATAAGCTGCTGGGCGTACGCGGTATCATCACCAGCAAAACCAAAAG BFGRP CCTGGATAAAGGCTACAATAAGACCTATCGCAGCCGCAAATATASCAGCTGGT ATCATCACCATCACCATCACTAATAACTCGAG-3′ (SEQ ID NO: 38) (ASC stands for AGC or ACC.) C430A-GAATTCTATAAGCTGCTGGCGGTACGCGGTATCATCACCAGCAAAACCAAAAG BFGRP CCTGGATAAAGGCTACAATAAGACCTATCGCAGCCGCAAATATASCAGCTGGT ATCATCACCATCACCATCACTAATAACTCGAG-3′ (SEQ ID NO: 39) (ASC stands for AGC or ACC.) C430S-GAATTCTATAAGCTGCTGAGCGTACGCGGTATCATCACCAGCAAAACCAAAAG BFGRP CCTGGATAAAGGCTACAATAAGACCTATCGCAGCCGCAAATATASCAGCTGGT ATCATCACCATCACCATCACTAATAACTCGAG-3′ (SEQ ID NO: 40) (ASC stands for AGC or ACC.) Belt GAATTCTATAAGCTGCTGTGTGTACGCGGTATCATCACCAGCAAAACCAAAAGCCT GGATAAAGGCTACAATAAGGCGCTGAACGATCTGTGCCATCACCATCACCATCAC TAATAACTCGAG (SEQ ID NO: 41)

Example 2. Expression of Non-Toxic Protease in *E. coli*

(83) Each of the expression vectors for *E. coli* constructed in Example 1 was transformed into an *E. coli* BL21 (DE3) strain as a host for gene expression to construct recombinant *E. coli* strains. LB culture medium was inoculated with the ampicillin-resistant colonies and culturing was performed with shaking at 37° C. When the absorbance at 600 nm reached 0.6 to 0.7, 0.5 mM IPTG was added to the culture medium, and then expression of the non-toxic protease was induced at 18 to 25° C. depending on the conditions. Next, the cells were cultured for 6 to 12 hours, and recombinant *E. coli* cells were harvested by centrifugation (at 3,000 rpm for 20 minutes).

(84) The harvested *E. coli* cells were suspended in a 25% sucrose solution (containing 50 mM Tris HCl, pH 7.8, 0.1 mM EDTA), and 0.2 mg/ml of lysozyme was added thereto, followed by incubation at 4° C. for 1.5 hours. Next, a 1.5-fold volume of lysis buffer (20 mM Tris HCl, pH7.8, 0.2 M NaCl, 1% DCA, 1.6% NP-40 (or 1% SDS), 2 mM EDTA, 0.5 mM DTT) was added to the suspension, and then the cells were lysed using ultrasound at 4° C. The protein was quantified and loaded on SDS-PAGE gel, followed by Coomassie blue staining.

(85) As a result, as shown in FIG. **7**, it could be confirmed that the wild-type and mutated non-toxic proteases were effectively expressed by induction with IPTG. As shown in FIG. **8**, it was confirmed that the non-toxic protease appeared only in the total cell lysate and the pellet, suggesting that it was mostly present in an insoluble form. In addition, as a result of subjecting the wild-type non-toxic protease to SDS-PAGE under non-reducing conditions and reducing conditions, it could be confirmed that the wild-type non-toxic protease existed in the form of high-molecular-weight protease under the non-reducing conditions, suggesting that disulfide bonds affect the formation of aggregates of the wild-type non-toxic protease (FIG. **9**). Meanwhile, when the wild-type and mutated non-toxic proteases were codon-optimized to be suitable for expression in *E. coli*, the expression levels thereof could be significantly increased (FIGS. **10** and **11**).

Example 3. Purification of Non-Toxic Protease

(86) To purify the non-toxic proteases, each of the recombinant *E. coli* strains was cultured according to the method of Example 2, and then the *E. coli* cells were harvested by centrifugation (at 3,000 rpm for 20 minutes). 1 g of the harvested cells were suspended in a cell suspending solution (50 mM Tris HCl, pH 7.8, 0.1 mM EDTA, 25% sucrose), and 0.2 mg/ml of lysozyme was added thereto, followed by incubation at 4° C. for 1.5 hours. Then, a 1.5-fold volume (3 ml) of lysis buffer (20 mM Tris HCl, pH 7.8, 0.2 M NaCl, 1% DCA, 1.6% NP-40 (or 1% SDS), 2 mM EDTA, 0.5 mM DTT) was added to the suspension, and the cells were disrupted using an ultrasonic disruptor until the viscosity of the cell lysate disappeared when viewed with the naked eye. The cell lysate was separated into a soluble fraction and an insoluble fraction by centrifugation at 12,000 rpm and 4° C. for 15 minutes.
(87) As a result of observing the fractions by Coomassie blue staining or Western blotting after SDS-PAGE, it was confirmed that a significant portion of each of the wild-type and mutated non-toxic proteases was present in the insoluble fractions.
(88) The non-toxic protease present in the soluble fraction was loaded onto a Ni.sup.2+-IDA affinity column (iminodiacetic acid). The column was washed with a 10-fold volume of binding buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) and a 6-fold volume of washing buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol+10 mM imidazole), and then eluted stepwise with elution buffers (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) containing 200, 300 or 500 mM imidazole.
(89) Meanwhile, after the soluble fraction was isolated, the non-toxic protease precipitate of the insoluble fraction was washed three times with 5 ml of the same volume as the supernatant (washing buffer, 2 to 4 M urea, 0.5% Triton X100, mM EDTA, 1 mM DTT) and the supernatant was removed. Meanwhile, after the soluble fraction was isolated, the non-toxic protease precipitate of the insoluble fraction was washed three times with 5 ml (the same volume as that of the supernatant) of washing buffer (2 to 4 M urea, 0.5% Triton X100, 1 mM EDTA, 1 mM DTT), and then the supernatant was removed. The recovered precipitate was denatured with 1 ml of urea solution (8 M urea, 20 mM Tris HCl, pH 7.8, 20 μM DTT), and then transferred to a dialysis membrane (molecular weight cutoff: 12,000, Sigma, cat no: D-0530, USA) and subjected to refolding by dialysis three times with a volume 100 times the volume of the urea solution of dialysis buffer (0.08 mM NaCl, 20 mM Tris HCl, pH 7.8, 0.03% tween 20), and 10 to 20 μM ZnCl.sub.2 was added thereto as needed. The resulting material was loaded onto a Ni.sup.2+-IDA affinity column (iminodiacetic acid), and the column was washed stepwise with a 10-fold volume of binding buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) and a 6-fold volume of washing buffer (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol+10 mM imidazole), and then eluted stepwise with elution buffers (50 mM potassium phosphate buffer (pH 8.0)+300 mM NaCl+1 mM PMSF+1 mM β-mercaptoethanol) containing each of 200, 300 and 500 mM imidazole. The fractions were combined and desalted with a PD-10 column (GE Healthcare, USA). Protein concentration was measured by BCA protein assay using bovine serum albumin as a standard.
(90) As a result, as shown in FIG. **14**, it could be confirmed that the wild-type and mutated non-toxic proteases and fusion non-toxic proteases were effectively purified.
Example 4. Analysis of Folding/Refolding Efficiencies of Non-Toxic Proteases
(91) As a result of performing the purification and refolding process on the insoluble fraction in Example 3, as shown in FIG. **15**, it was confirmed that, in the case of the wild-type protease, aggregation still appeared after the refolding performed by dialysis, but in the case of the mutated non-toxic protease or the mutated fusion non-toxic protease, aggregation hardly appeared after refolding.
(92) Meanwhile, after centrifugation at 12,000 rpm for 15 minutes, the content (%) of the non-toxic

protease in each of the soluble fraction and the insoluble fraction (pellet fraction, aggregation) was analyzed by Coomassie blue staining after SDS-PAGE. As a result, as shown in FIG. **16**, it was confirmed that about 60 to 90% of each of the wild-type and mutated non-toxic proteases was present in the insoluble fraction, suggesting that each of the non-toxic proteases was mostly present in the insoluble fraction rather than the soluble fraction. However, the tendency of the non-toxic-protease to be recovered through the refolding process was completely different between the wild-type non-toxic protease and the mutated non-toxic protease. That is, the wild-type non-toxic-protease was mostly present in a precipitated inclusion body state even after the refolding process, and thus was impossible to recover, whereas almost all of the mutated non-toxic-protease could be refolded and recovered.
Example 5. Analysis of Enzymatic Activity of Non-Toxic Protease
(93) In order to analyze the enzymatic activities of the non-toxic proteases and variants thereof according to the present invention, a substrate for analyzing the cleavage by the non-toxic protease was synthesized. As the substrate, the ANNEXIN V protein fused to the C-terminus of SNAP25 was used for expression (Table 3), and a nucleotide sequence encoding the SNAP25-ANNEXIN V fusion protein was cloned into a pET22b (+) expression vector which was then transformed into an *E. coli* BL21 (DE3) strain. The transformed *E. coli* strain was cultured with shaking in LB medium at 37° C., and when it reached an O.D.sub.600 of 0.6 to 0.7, 0.5 to 1 mM IPTG was added thereto, and the strain was cultured for 6 hours to induce protein expression. After the cells were disrupted with an ultrasonic disruptor, only the soluble fraction (supernatant) was recovered and purification was performed using his-6 tag contained in SNAP25-ANNEXIN V in a manner similar to the method of Example 3. As a result, as shown in FIG. **17**, it was confirmed that SNAP25-ANNEXIN V was effectively expressed and purified.
(94) In order to confirm whether all of the non-toxic proteases purified according to the present invention retain the same enzymatic activity as the wild-type non-toxic protease, cleavage of the purified non-toxic proteases was tested using the purified SANP25-ANNEXIN V as a substrate. To this end, the prepared SNAP25-ANNEXiN V and the non-toxic protease were mixed together at a ratio of 20:1 (w/w) (substrate (SNAP25-ANNEXINV): enzyme (protease)) in an enzymatic reaction solution (100 mM HEPES pH 7.4, 1 mM NaCl, 20 μM ZnCl.sub.2, mM DTT 2) and subjected to enzymatic reaction at 37° C. for 5 hours. As a result of analyzing the samples after completion of the reaction, it was confirmed that all of the non-toxic proteases according to the present invention cleaved 50-kDa SNAP25-ANNEXiN V to form 36-kDa and 14-kDa protein bands. This demonstrated that the mutated non-toxic proteases with improved productivity maintained their enzyme activity without changes.
(95) TABLE-US-00003 TABLE 3 Amino acid sequence Nucleotide sequence His-SNAP25-MGGSHHHHHHENLYFQGSGGNKLKSSD 5′- ANNEXIN V
AYKKAWGNNQDGVVASQPARVVDERE catATGGGCGGCAGCCATCATCATCATCA
QMAISGGFIRRVTNDARENEMDENLEQV TCATGAAAACCTGTATTTTCAGGGCTCT
SGIIGNLRHMALDMGNEIDTQNRQIDRIM GGCGGCAACAAGCTGAAATCTAGCGAT
EKADSNKTRIDEANQRATKMLGSGAQV GCTTACAAAAAAGCCTGGGGCAATAATC
LRGTVTDFPGFDERADAETLRKAMKGL AGGACGGCGTGGTGGCCAGCCAGCCTGC
GTDEESILTLLTSRSNAQRQEISAAFKTLF TCGTGTAGTGGACGAACGTGAGCAGATG
GRDLLDDLKSELTGKFEKLIVALMKPSR GCCATCAGCGGCGGCTTCATCCGTCGTG
LYDAYELKHALKGAGTNEKVLTEIIASR TAACCAATGATGCCCGTGAAAATGAAAT
TPEELRAIKQVYEEEYGSSLEDDVVGDTS GGATGAAAACCTGGAGCAGGTGAGCGG
GYYQRMLVVLLQANRDPDAGIDEAQVE CATCATCGGCAACCTGCGTCACATGGCC
QDAQALFQAGELKWGTDEEKFITIFGTRS CTGGATATGGGCAATGAGATCGATACCC
VSHLRKVEDKYMTISGFQIEETIDRETSG AGAATCGTCAGATCGACCGTATCATGGA
NLEQLLLAVVKSIRSIPAYLAETLYYAMK GAAGGCTGATTCCAACAAAACCCGTATC
GAGTDDHTLIRVMVSRSEIDLENIRKEFR GATGAGGCCAACCAACGTGCAACCAAG

KNFATSLYSMIKGDTSGDYKKALLLLCG ATGCTGGGCAGCGGCGCACAGGTTCTGC EDD (SEQ ID NO: 42) GTGGCACTGTGACCGACTTCCCTGGCTT TGATGAGCGTGCTGATGCAGAAACCCTG CGTAAGGCTATGAAAGGCCTGGGCACCG ATGAGGAGAGCATCCTGACCCTGCTGAC CTCCCGTAGCAATGCTCAGCGTCAGGAA ATCTCTGCAGCTTTTAAGACCCTGTTTGG CCGTGATCTGCTGGATGACCTGAAATCC GAACTGACCGGCAAATTTGAAAAACTGA TCGTGGCTCTGATGAAACCTTCTCGTCTG TATGATGCTTATGAACTGAAACATGCCC TGAAGGGCGCTGGCACCAATGAAAAAG TACTGACCGAAATCATTGCTTCTCGTAC CCCTGAAGAACTGCGTGCCATCAAACAA GTTTATGAAGAAGAATATGGCTCTAGCC TGGAAGATGACGTGGTGGGCGACACTTC TGGCTACTACCAGCGTATGCTGGTGGTT CTGCTGCAGGCTAACCGTGACCCTGATG CTGGCATCGATGAAGCTCAAGTTGAACA AGATGCTCAGGCTCTGTTTCAGGCTGGC GAACTGAAATGGGGCACCGATGAAGAA AAGTTTATCACCATCTTTGGCACCCGTA GCGTGTCTCATCTGAGAAAGGTGTTTGA CAAGTACATGACCATCTCTGGCTTTCAA ATCGAGGAAACCATCGACCGTGAGACTT CTGGCAATCTGGAGCAACTGCTGCTGGC TGTTGTGAAATCTATCCGTAGCATCCCT GCCTACCTGGCAGAGACCCTGTATTATG CTATGAAGGGCGCTGGCACCGATGATCA TACCCTGATCCGTGTCATGGTTTCCCGTA GCGAGATCGATCTGTTTAACATCCGTAA GGAGTTTCGTAAGAATTTTGCCACCTCT CTGTATTCCATGATCAAGGGCGATACCT CTGGCGACTATAAGAAAGCTCTGCTGCT GCTGTGTGGCGAAGATGACTAActcgag-3′ (SEQ ID NO: 43)

Example 6. EXPRESSION OF NON-TOXIC PROTEASE IN YEAST

(96) In another embodiment, to express a non-toxic protease in yeast, the sequence shown in Table 4 below was subcloned by restriction enzymes (BamHI and NotI) downstream of the AOX1 promoter of pPIC9. Meanwhile, his 6-tag was introduced for purification of the non-toxic protease, and a TEV cleavage enzyme peptide sequence was added in order to remove his 6-tag after purification, thereby constructing a vector for expression in *Pichia pastoris.*

(97) A *Pichia pastoris* strain was transformed with the vector and cultured in histidine-deficient medium. Then, the formed colonies were cultured in the methanol-assimilating yeast medium BM (buffered minimal medium: 100 mM potassium phosphate, pH 6.0 1.34% yeast nitrogen base, 4×10.sup.−5% biotin) by supplying glycerol (1%) as a carbon source, and then the carbon source was replaced with methanol (0.5%), thus inducing expression of the non-toxic protease. The culture medium was recovered, and the expressed non-toxic protease was purified therefrom using His-Tag and subjected to Western blotting with a botulinum light-chain antibody (cat no. PA5-48053 (ThermoFisher Scientific, USA). Meanwhile, since a non-toxic protease with an increased molecular weight due to glycosylation is expressed in yeast, the non-target protease was aglycosylated by reaction with the enzyme PNGaseF (cat no. P0704S, New England Bio Lab USA), which cuts sugar chains from the protein, according to the manufacturer's instructions. Then, the non-target protease was subjected to SDS-PAGE and then to Western blotting with a botulinum light-chain antibody.

(98) As a result, as shown in FIG. **19**, it was confirmed that the non-toxic protease was effectively expressed in and purified from the yeast, and a multi-band, non-toxic protease with increased molecular weight due to glycosylation was identified in the yeast, and a single band of the non-toxic protease was observed after aglycosylation.

(99) TABLE-US-00004 TABLE 4 Amino acid sequence Nucleotide sequence SMF-
MRFPSIFTAVLFAASSALAAPVNTTTED 5- (ΔM)BTX-
ETAQIPAEAVIGYSDLEGDFDVAVLPFS atgagatttccatctatttttactgcagttttgtttgcagcatcttctgca L-
PEP1- NSTNNGLLFINTTIASIAAKEEGVSLEK
ttggcagcaccagttaacactactactgaagatgaaactgcacaaattcc TEV-His
RQFVNKQFNYKDPVNGVDIAYIKIPNV agcagaagcagttattggttactctgatttggaaggtgattttgatgttg
GQMQPVKAFKIHNKIWVIPERDTFTNP ctgttttgccattttctaactctactaataacggtttgttgtttattaata
EEGDLNPPPEAKQVPVSYYDSTYLSTD ctactattgcatctattgcagcaaaggaagaaggtgtttctttggaaaaaa

NEKDNYLKGVTKLFERIYSTDLGRMLL gacagtttgttaacaagcaattcaattacaaagatccagttaatggtgttg TSIVRGIPFWGGSTIDTELKVIDTNCINV acattgcttatattaaaattccaaacgttggtcagatgcagccagttaaa IQPDGSYRSEELNLVIIGPSADIIQFECKS gctttcaagatccataacaaaatttgggttatcccagagagagacactt FGHEVLNLTRNGYGSTQYIRFSPDFTEG tcactaatccagaggaaggtgatttgaacccaccaccagaagcaaaa FEESLEVDTNPLLGAGKFATDPAVTLA caggttccagttagttattatgacagtacttatctttccactgataacgag HELIHAGHRLYGIAINPNRVFKVNTNA aaagataactatttgaaaggtgttactaagctgtttgaaagaatttacag YYEMSGLEVSFEELRTFGGHDAKFIDS tactgacttgggaagaatgttgctgacttcaattgttagaggaattccat LQENEFRLYYYNKFKDIASTLNKAKSI tttggggtggatctactattgacactgaattgaaagttatcgatactaatt VGTTASLQYMKNVFKEKYLLSEDTSG gtattaacgttatccaaccagacggttcctacagatctgaggagcttaa KFSVDKLKFDKLYKMLTEIYTEDNFVK cttggttattattggtccatccgctgacattattcaatttgagtgtaagtc FFKVLNRKTYLNFDKAVFKINIVPKVN attcggacatgaggttttgaacctgactcgtaatggttatggttccactca YTIYDGFNLRNTNLAANFNGQNTEINN atatattagattttccccagattttacttttggtttcgaggaaagtttg MNFTKLKNFTGLFEFYKLLCVRGIITSK gaggttgatactaacccattgctgggtgcaggtaagtttgctactgatcca TKSLDKGYNKKETWWETWWTEW gcagttactcttgctcatgagctgatccatgcaggtcatagattgtatggt SQPKKKRKVENLYFQSNHHHHHH attgcaattaatccaaaccgtgtttttaaagttaatactaatgcttattat (SEQ ID NO: 44) gaaatgtcaggtttggaagtttctttcgaagagttgagaactttcggagg acatgatgctaagttcattgactctttgcaggaaaacgagttccgtctgt actactataacaagttcaaggacatcgcatctactttgaacaaggcaa agtcaattgttggtactactgcttcactgcaatatatgaagaacgttttca aggagaagtacttgttgtccgaggatactagtggtaaattctctgttga caaattgaagttcgacaaattgtacaaaatgctgactgagatctatact gaagacaacttcgttaagtttttttaaagttctgaacagaaagacttatttg aactttgataaggctgtttttaagattaacattgttccaaaggttaactat actatttatgacggtttcaatctgagaaacactaatcttgcagctaatttc aatggtcagaatactgagattaataatatgaacttcactaagttgaaaaat tttactggtttgtttgaattttacaagttgttgtgtgttcgtggaattatt acttccaagactaaatctttggataagggttacaacaaaaaggaaacttgg tgggaaacttggtggactgaatggtctcaaccaaagaagaagagaa aggttgagaacctg tacttccaatccaatcatcaccatcaccatcacta a-3′ (SEQ ID NO: 45) SMF MRFPSIFTAVLFAASSALAAPVNTTTED 5′- ETAQIPAEAVIGYSDLEGDFDVAVLPFS atgagatttccatctatttttactgcagttttgtttgcagcatcttctgca NSTNNGLLFINTTIASIAAKEEGVSLEK ttggcagcaccagttaacactactactgaagatgaaactgcacaaattcc R (SEQ ID NO: 46) agcagaagcagttattggttactctgatttggaaggtgattttgatgttg ctgttttgccattttctaactctactaataacggtttgttgtttattaata ctactattgcatctattgcagcaaaggaagaaggtgttTctttggaaaaaa ga-3′ (SEQ ID NO: 47) TEV ENLYFQSN (SEQ ID NO: 48) 5′-gagaacctgtacttccaatccaat-3′ (SEQ ID NO: 49)

Example 7. Massive Fermentation and Purification of Non-Toxic Protease

(100) It was attempted to ferment and purify large amounts of the recombinant *E. coli* strains constructed in Example 2. To this end, the stock solution of each recombinant strain was inoculated into LB medium supplemented with ampicillin and was cultured with shaking overnight at 37° C. and 200 rpm (seed culture). 3 to 4 L of LB medium supplemented with 50 μg/ml of ampicillin was placed in a 7-L jar fermenter, and 50 ppm of an Antifoam B emulsion was added thereto. Then, 30 mL of the strain cultured overnight was inoculated into the medium and fermented at 37° C. at an impeller speed of 250 rpm under an air pressure corresponding to 2 L/min of filtered air. When the strain reached an OD.sub.600 of 0.3, the internal temperature of the fermenter was lowered with a cooler of the fermenter and the strain was cultured. When the strain reached an OD.sub.600 of 0.6 to 0.7, 0.5 mM IPTG was added and the strain was cultured overnight (16 to 18 hours) at 18° C.

(101) 200 ml of lysis buffer (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol) was added to the cell pellet (about 5 g by wet weight) of 1L of the fermentation culture medium, and the cells were completely suspended. A buffer for cell disruption (1 mM PMSF, 1 mM β-ME (beta-mercaptoethanol), 0.1% Triton-X 100) was added to the cell suspension, and the cells were sonicated five times using a Sonics Vibra-cell sonicator for 4 min at pulse on/off of 4 sec/8 sec and

at power of 65%. The cell lysate was divided into a supernatant and a cell lysate precipitate by centrifugation at 4° C. at 12,000 rpm for 30 minutes.

(102) Meanwhile, a Ni-IDA column (Workbeads™ 40 Ni-IDA, 40-650-001, Bio-Works, Sweden) was packed and the column was equilibrated with a 10-fold volume of equilibration buffer (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol, 0.1% TritonX-100, 1 mM PMSF, 1 mM β-ME). Ni-IDA resin was added to the centrifuged supernatant containing the soluble protein to induce a binding reaction with his-tag of the non-toxic protease at 4° C. for 2 hours, and was loaded into the column. The column was washed with a 20-fold volume of washing buffer A (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol, 1 mM β-ME), and then washed with a 20-fold volume of washing buffer (50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 50 mM imidazole, 10% glycerol, 1 mM β-ME). The column was eluted stepwise with 1.5-fold volumes of elution buffers (100/200/300/500/700 mM imidazole, 50 mM Tris-HCl (pH 8.0), 150 mM NaCl, 10% glycerol, 1 mM β-ME). EDTA was added to each eluted fraction to a final concentration of 1 mM. The purity of each eluted fraction was analyzed by SDS-PAGE, and the high-purity eluted fractions obtained using the elution buffers containing 500 to 700 mM imidazole were combined, transferred to a dialysis membrane (molecular weight cut-off: 10-30 kDa), and dialyzed using PBS buffer to remove other chemical components, and the purified non-toxic protease was recovered. For cryopreservation, the non-toxic protease was dialyzed with PBS buffer containing 10% glycerol (FIG. **20**).

(103) As a result, as shown in FIGS. **21** to **26**, it was confirmed that the wild-type non-toxic protease was recovered from the soluble fraction at a concentration of 3 to 6 mg/L, the non-toxic protease obtained by substituting the amino acid cysteine at position 430 of the wild-type non-toxic protease with serine was recovered from the soluble fraction at a concentration of 8 to 12 mg/L, and the non-toxic protease obtained by substituting the amino acid cysteine at position 430 of the wild-type non-toxic protease with alanine was recovered from the soluble fraction at a concentration of 15 to 20 mg/L, whereas the non-toxic protease obtained by substituting the amino acid cysteine at position 430 of the wild-type non-toxic protease with glycine was recovered from the soluble fraction at a concentration of 40 to 60 mg/L, suggesting that cysteine-to-glycine substitution at position 430 significantly increased the recovery of the non-toxic protease from the soluble fraction. It was believed that the reason why the recovery of the non-toxic protease from the soluble fraction can be enhanced as described is because cysteine-to-glycine substitution at position 430 of the non-toxic protease inhibited the formation of a bisulfide bond, the amount of the *E. coli* strain was increased through culturing at 37° C., the metabolism of the *E. coli* strain was slowed by lowering the culture temperature to 18° C. when an appropriate O.D. was reached, and expression of the protein was induced relatively slowly by adding IPTG at a decreased concentration of 0.5 mM, whereby the refolding efficiency of the non-toxic protease was increased. Meanwhile, in the case in which cysteine at position 430 was substituted with alanine or serine, a non-specific disulfide bond could be induced, and thus the efficiency of production of the non-toxic proteases from the soluble fraction was lower than that in the case in which cysteine at position 430 was substituted with glycine.

(104) This effect of increasing the efficiency of expression purification of the mutated non-toxic protease in and from the soluble protein was higher than that for the wild-type non-toxic protease, even when the BFGFRP domain of the fibroblastic growth factor was fused to the mutated non-toxic protease. Thus, it could be seen that, even when another peptide is fused to the non-toxic peptide to the mutated non-toxic protease, the efficiency of expression and efficiency of purification of the mutated non-toxic protease is maintained. Specifically, the fusion non-toxic protease obtained by substituting cysteine at position of the non-toxic protease with glycine and fusing BFGFRP to the non-toxic protease was recovered at a concentration of 15 to 80 mg/L.

Example 8. Examination of Single Dose Toxicity of Non-Toxic Proteases

(105) In order to confirm whether the toxicity of the non-toxic protease to be used in the present

invention is reduced, the single dose toxicity thereof was evaluated in mice. An experiment was conducted using the wild-type non-toxic protease represented by SEQ ID NO: 1 of the present invention, and a “test for single dose administration of intraperitoneal administration of test substance in mice” was performed by the animal testing institution KPC (212-9, Opo-ro, Opo-eup, Gwangju-si, Gyeonggi-do, Korea).
(106) As experimental animals, ICR mice were purchased from ORIENTBIO Inc. (Korea). The experiment was carried out using 35 6-week-old male mice, and animals that were not abnormal in appearance were brought into the breeding area and acclimatized in the animal room where the test was conducted, for 7 days. After 7 days of quarantine and acclimatization, healthy animals were selected by checking their health and suitability for testing. For drinking water, tap water was sterilized with a filter water sterilizer, irradiated with UV light, and freely fed using a drinking water bottle (250 mL) made of polycarbonate. The animals were allowed to freely access the rodent feed for laboratory animals provided by Cargill Agri Purina, Inc.
(107) TABLE-US-00005 TABLE 5 Number Administration Test of route and Observation group Drug Dose (mg/kg) Volume animals frequency period G1 Vehicle n/a 0.25 mL G2 Test 1 μg/mouse 0.25 Ml Five mice Intraperitoneal 4 days after substance per group injection, administration G4 Test 10 μg/mouse 0.25 mL single substance G4 Test 40 μg/mouse 0.25 mL substance G5 Test 100 μg/mouse 0.25 mL substance G6 Test 110 μg/mouse 0.25 mL substance G7 Test 120 μg/mouse 0.25 mL substance
(108) As shown in Table 5 above, as the test substance, the wild-type non-toxic protease from which the heavy chain has been removed was purified according to the present invention and intraperitoneally injected into ICR mice (average body weight 34 g, n=5 for each concentration) at a dose of 1 to 120 μg/mouse. As a result, as shown in Table 6, it was confirmed that all the mice survived. Here, the dose (120 μg/mouse) for test group G7 corresponds to about 3.6 mg/kg mouse body weight. It is known that the median lethal dose (LD50) of natural botulinum toxin type A is about 0.3 ng/kg in rodent mice, and about 1 ng/kg in humans, which corresponds to a lethal dose of 1 μg for a 70-kg adult (Annu. Rev. Microbiol. 1999. 53:551-75, Eric A. Johnson, CLOSTRIDIAL TOXINS AS THERAPEUTIC AGENTS: Benefits of Nature's Most Toxic Proteins). Therefore, it was confirmed that the toxicity of the wild-type non-toxic protease used in the present invention decreased by 1.2×10.sup.7 times compared to the reported median lethal dose (LD50) (0.3 ng/kg) of the natural botulinum toxin type A in mice, and that the mutated non-toxic protease of the present invention also had a significantly reduced median lethal dose (LD50) in mice, which is similar to that of the wild-type non-toxic protease.
(109) TABLE-US-00006 TABLE 6 Summary: Incidence of mortality and daily observations Clinical Days Group Sex N Mortality observation 1 2 3 4 5 G1 Male 5 0% No clinical 5 5 5 5 5 (Vehicle) sign G2 Male 5 0% No clinical 5 5 5 5 5 (1 μg/mouse) sign G3 Male 5 0% No clinical 5 5 5 5 5 (10 μg/mouse) sign G4 Male 5 0% No clinical 5 5 5 5 5 (50 μg/mouse) sign G5 Male 5 0% No clinical 5 5 5 5 5 (100 μg/mouse) sign G6 Male 5 0% No clinical 5 5 5 5 5 (110 μg/mouse) sign G7 Male 5 0% No clinical 5 5 5 5 5 (120 μg/mouse) sign
(110) Although the present invention has been described above with reference to the embodiments, it is to be understood that the present invention is not necessarily limited to the embodiments, and various modifications are possible without departing from the scope and spirit of the present invention. Accordingly, the scope of the present invention should be construed to include all embodiments falling within the scope of the claims appended hereto.
INDUSTRIAL APPLICABILITY
(111) The present invention has advantages in that it is possible to inhibit aggregation into inclusion bodies, which generally occurs in the process of producing wild-type non-toxic protease using a recombinant microorganism, and thus it is possible to obtain active mutated non-toxic proteases from both a soluble fraction and an insoluble fraction, which are formed by disrupting the recombinant microorganism, even by a simple dialysis/refolding process alone, and accordingly, it

is possible to produce non-toxic proteases in high yield.
SEQUENCE LIST FREE TEXT
(112) An electronic file is attached.

Claims

1. A mutated non-toxic protease in which an amino acid cysteine (Cys) at position 430 of a non-toxic protease represented by the amino acid sequence of SEQ ID NO: 1 is substituted with an amino acid glycine (Gly).
2. A fusion non-toxic protease in which the mutated non-toxic protease of claim 1 is fused with any one or more peptides selected from the group consisting of: i) a cell penetrating peptide; ii) a belt domain fragment peptide; and iii) a cell targeting peptide.
3. The fusion non-toxic proteinase of claim 2, wherein the cell penetrating peptide is represented by the amino acid sequence of SEQ ID NO: 3, the belt domain fragment peptide is represented by the amino acid sequence of SEQ ID NO: 5, and the cell targeting peptide is represented by the amino acid sequence of SEQ ID NO: 7.
4. The fusion non-toxic proteinase of claim 2, wherein the any one peptide is fused directly or via a linker to the non-toxic protease.
5. The fusion non-toxic proteinase of claim 4, wherein the linker is a peptide linker represented by (Gly) N wherein N is an integer ranging from 3 to 10.
6. The fusion non-toxic proteinase of claim 2, further comprising a cleavable peptide sequence by a protease for cleavage between a C-terminus of the non-toxic protease and an N-terminus of the peptide fused with the non-toxic protease.
7. The fusion non-toxic proteinase of claim 6, wherein the cleavable peptide sequence is represented by the amino acid sequence of SEQ ID NO: 27.
8. A mutant gene encoding the mutated non-toxic protease of claim 1.
9. The mutant gene of claim 8, which is represented by a nucleotide sequence of SEQ ID NO: 30.
10. A gene construct comprising: the mutant gene of claim 9; and any one or more nucleic acids selected from the group consisting of i) a nucleic acid encoding a cell penetrating peptide; ii) a nucleic acid encoding a belt domain fragment peptide; and iii) a nucleic acid encoding a cell targeting peptide.
11. The gene construct of claim 10, wherein the nucleic acid encoding the cell penetrating peptide is represented by the nucleotide sequence of SEQ ID NO: 4, the nucleic acid encoding the belt domain fragment peptide is represented by the nucleotide sequence of SEQ ID NO: 6, and the nucleic acid encoding the cell targeting peptide is represented by the nucleotide sequence of SEQ ID NO: 8.
12. The gene construct of claim 10, further comprising a nucleic acid encoding a linker peptide between the mutant gene encoding the mutated non-toxic protease and the any one or more nucleic acids.
13. The gene construct of claim 12, wherein the nucleic acid encoding the linker peptide is represented by the nucleotide sequence of SEQ ID NO: 26.
14. The gene construct of claim 10, further comprising a nucleic acid encoding a peptide sequence cleavable by a cleavage protease between the mutant gene encoding the mutated non-toxic protease and the one or more nucleic acids.
15. A pharmaceutical composition for treating Dystonia containing the non-toxic protease of claim 1 as an active ingredient.
16. The pharmaceutical composition of claim 15, wherein the Dystonia is selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis.

17. The pharmaceutical composition of claim 16, which is for transdermal administration.
18. A hyaluronic acid microneedle patch containing the non-toxic protease of claim 1.
19. The hyaluronic acid microneedle patch of claim 18, which is for treatment or alleviation of a symptom selected from the group consisting of facial spasm, eyelid spasm, torticollis, blepharospasm, spasmodic torticollis, cervical dystonia, oromandibular dystonia, spasmodic dysphonia, migraine, anal pruritus, and hyperhidrosis.